

Yuexin ZHANG

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Employment

Center for Astrophysics, Harvard & Smithsonian, US	Nov 2023 – Present
NWO Rubicon Fellow (host institution; Nov 2023 – Oct 2025)	
Astrophysicist (from Oct 2025)	
Kapteyn Astro. Institute, University of Groningen, NL	Oct 2023 – Oct 2025
NWO Rubicon Fellow (home institution)	

Education

Kapteyn Astro. Institute, University of Groningen, NL	Oct 2019 – Sep 2023
PhD in Astronomy. Advisor: Mariano Méndez and Diego Altamirano	
Department of Physics, Fudan University, CN	Sep 2015 – June 2019
B.S. (<i>Honor</i>) in Physics. Advisor: Cosimo Bambi	
Hamburg University, DE	Jul 2018
Summer Exchange Student	

Previous Academic Positions

University of Southampton, UK	Feb 2023
Visiting Researcher. PI: Diego Altamirano	
Institute of High-Energy Physics, CAS, CN	Mar 2021 – Mar 2022
Visiting Researcher. PI: Shuang-Nan Zhang and Jin-Lu Qu	
Shanghai Astronomical Observatory, CAS, CN	Jun 2019 – Sep 2019
Summer Visiting Student. PI: Wenfei Yu	

Member

<i>IXPE</i> Science Team	2024 – Present
<i>NICER</i> Science Team	2023 – Present
<i>Insight</i> -HXMT Science Team	2021 – 2022

Professional Service

Reviewer for Nature Astronomy, ApJ
Panelist for NASA proposals

2025 – Present
2024 – 2025

Awards

NWO Rubicon Fellowship (153k EUR)	2023 – 2025
<i>“New light on the origin of X-ray variability in Galactic accreting black holes”</i>	
CSC and UoG Scholarship (96k EUR)	2019 – 2023
<i>“Spectral variability of compact X-ray sources”</i>	
Shanghai Outstanding Graduate	2019
Wangdao Scholar (named after the former president of Fudan)	2019
Fudan’s Undergraduate Research Opportunities Program (10k CNY)	2018 – 2019
<i>“Testing general relativity using X-ray reflection spectroscopy”</i>	

Observations

PI:

- [1] *Chandra* Cycle 27, “A multi-wavelength, multi-dimensional campaign of the disk wind in *GRS 1915+105*”, 50 ks (62k USD)
- [2] *Magellan* 2025B, “The Mass and Spin of Two Black Hole X-ray Binaries”, 3 half-nights
- [3] *NuSTAR* Cycle 11, “Can power color hue efficiently capture the next transition in *GX 339-4* with *NuSTAR-NICER?*”, 60 ks (74k USD)
- [4] *NuSTAR* Cycle 11, “A multi-wavelength, multi-dimensional view of *GRS 1915+105* in its wind phase”, 40 ks (60k USD)
- [5] *NICER* Cycle 7, “A multi-wavelength, multi-dimensional view of the accretion wind in *GRS 1915+105*”, 50 ks (39k USD)
- [6] *NICER* DDT on V4641 Sgr, 30 ks
- [7] *Magellan* 2024B, “The mass and spin of *Swift J1727.8–1613*”, 4 half-nights
- [8] *NICER* DDT on *GX 339–4*, 10 ks
- [9] *Insight-HXMT* AO–6, “Unravelling the origin of fast time variability in X-ray binaries with *Insight-HXMT*”, 3.6 ms
- [10] *Insight-HXMT* AO–5, “On the QPO properties of a black-hole X-ray binary”, 10 ks

Co-I:

- [1] *NuSTAR* Cycle 11, “Observation of a black hole X-ray binary in the dim hard state”, PI: König, 100 ks
- [2] *NICER* Cycle 7, “A legacy of turtles in *NICER* Cycle 7: Amassing a comprehensive spectral-timing archive of outbursting black holes”, PI: Steiner, 120 ks
- [3] *NICER* Cycle 7, “A long look at the hard state of *LMC X-3* with *NICER*”, PI: Steiner,

30 ks

- [4] *NICER* Cycle 7, “Monitoring a transient black hole X-ray binary in the soft-to-hard transition”, PI: König, 60 ks
- [5] *IXPE* Cycle 2, “A test of disk-jet misalignment and black hole spin measurement for the X-ray binary *H1743–322*”, PI: Steiner, 320 ks
- [6] *IXPE* Cycle 2, “First polarimetric look into the heartbeat state of accreting black holes with *4U 1630–47*”, PI: Steiner, 300 ks
- [7] *NuSTAR* Cycle 10, “Slim-disk spectro-polarimetry with *LMC X-3*”, PI: Steiner, 120 ks
- [8] *IXPE* Cycle 1, “Evolving slim-disk polarization with *LMC X-3*”, PI: Steiner, 1.2 ms

Conferences and Talks

- [1] Conference, 22nd Meeting of the High Energy Astrophysics Division, St. Louis, US12–16 Oct 2025
“A *NICER* view of the corona through time-dependent Comptonization of the quasi-periodic oscillations in nine black-hole X-ray binaries (Poster)”
- [2] Conference, 21st Meeting of the High Energy Astrophysics Division, Horseshoe Bay, US 7–12 Apr 2024
“The evolution of the high-frequency variability in the black hole candidate *GRS 1915+105* as seen by *RXTE* (Poster)”
- [3] Seminar, MIT, USA 21 Feb 2024
“A *NICER* view of the corona through Comptonization modeling of the QPO in bright black-hole X-ray binaries”
- [4] Seminar, Wuhan University, CN 8 Oct 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [5] Seminar, University of Amsterdam, NL 13 Sep 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [6] Colloquium, Yunnan Observatory, CN 14 Aug 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [7] Seminar, Fudan University, CN 1 Aug 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [8] Seminar, Institute of High-Energy Physics, CN 27 Jul 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [9] Conference, The First Vasto Accretion Meeting, Vasto, IT 19–23 Jun 2023
“The corona-jet evolution of black-hole X-ray transients *MAXI J1535–571* and *MAXI J1820+070* (Poster)”
- [10] Conference, 10th Microquasar Workshop, Crete, GR 22–26 May 2023

- “The jet-like corona of black-hole X-ray transient from the HIMS to the SIMS”*
- [11] Conference, Nederlandse Astronomen Conferentie 2023, Leeuwarden, NL 15–17 May 2023
“Corona as the energy reservoir for radio jet: a case study of GRS 1915+105 (Poster)”
- [12] Conference, NOVA Network NW3, Amsterdam, NL 19 Jan 2023
“The jet-like corona of black-hole X-ray transients from the HIMS to the SIMS”
- [13] Conference, 44th COSPAR Scientific Assembly, Athens, GR 16–24 Jul 2022
“Corona evolution of MAXI J1535–571 revealed by type-C quasi-periodic oscillations observed with Insight-HXMT”
“The evolution of the high-frequency variability in GRS 1915+105 as seen by RXTE (Poster)”
- [14] Conference, Black Hole Accretion under the X-ray Microscope, Madrid, ES 14–17 Jun 2022
“The evolution of the corona in MAXI J1535–571 through type-C quasi-periodic oscillations with Insight-HXMT (Poster)”
- [15] Conference, China Astronomy Annual Meeting (online), Nanchong, CN 2–6 Dec 2021
“Mapping the Comptonization region of black holes up to 100 keV through quasi-periodic oscillations in the intermediate state with Insight-HXMT”

References

Prof. Mariano Méndez: mariano@astro.rug.nl
 Dr. James (“Jack”) Steiner: james.steiner@cfa.harvard.edu
 Prof. Diego Altamirano: d.altamirano@soton.ac.uk
 Prof. Shuang-Nan Zhang: zhangsn@ihep.ac.cn
 Prof. Cosimo Bambi: bambi@fudan.edu.cn

Publications

First and second author:

- [1] **Zhang, Y.**, Méndez, M., Motta, S. et al. (2024). *A systematic study of the high-frequency bump in the black-hole low-mass X-ray binary GX 339–4*. Monthly Notices of the Royal Astronomical Society, 527(3), 5638–5648.
- [2] **Zhang, Y.**, Méndez, M., García, F. et al. (2023). *A NICER look at the jet-like corona of MAXI J1535–571 through type-B quasi-periodic oscillation*. Monthly Notices of the Royal Astronomical Society, 520(4), 5144–5156.
- [3] **Zhang, Y.**, Méndez, M., García, F. et al. (2022). *The evolution of the high-frequency variability in the black hole candidate GRS 1915+105 as seen by RXTE*. Monthly Notices of the Royal Astronomical Society, 514(2), 2891–2901.

- [4] **Zhang, Y.**, Méndez, M., García, F. et al. (2022). *The evolution of the corona in MAXI J1535-571 through type-C quasi-periodic oscillations with Insight-HXMT*. Monthly Notices of the Royal Astronomical Society, 512(2), 2686-2696.
- [5] Tripathi, A., **Zhang, Y.**, Abdikamalov, A. B. et al. (2021). *Testing general relativity with NuSTAR data of galactic black holes*. The Astrophysical Journal, 913(2), 79.
- [6] **Zhang, Y.**, Abdikamalov, A. B., Ayzenberg, D. et al. (2019). *Tests of the Kerr hypothesis with GRS 1915+105 using different RELXILL flavors*. The Astrophysical Journal, 884(2), 147.
- [7] **Zhang, Y.**, Abdikamalov, A., Ayzenberg, D. et al. (2019). *About the Kerr nature of the stellar-mass black hole in GRS 1915+105*. The Astrophysical Journal, 875 (1), 41.
- [8] **Zhang, Y.**, Zhou, M., & Bambi, C. (2018). *Iron line spectroscopy of black holes in asymptotically safe gravity*. The European Physical Journal C, 78 (5), 376.

Co-author:

- [1] Fan, N., Steiner, J., Bambi, C. et al., **incl. Zhang, Y.** (2024). *NICER spectral and timing analysis of 4U 1630–47 and its heartbeat state*. The Astrophysical Journal, 984(1), 31.
- [2] Alabarta, K., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2024). *Geometry of the Comptonization region of MAXI J1348–630 through type-C quasi-periodic oscillations with NICER*. The Astrophysical Journal, 980(2), 251.
- [3] Zhang, Z., Bambi, C., Liu, H. et al., **incl. Zhang, Y.** (2024). *A variable ionized disk wind in MAXI J1803–298 revealed by NICER*. The Astrophysical Journal, 975(1), 22.
- [4] Podgorny, J., Svoboda, J., Dovciak, M. et al., **incl. Zhang, Y.** (2024). *Recovery of the X-ray polarisation of Swift J1727.8–1613 after the soft-to-hard spectral transition*. Astronomy & Astrophysics, 686, L12.
- [5] Jin, P., Zhang, G., **Zhang, Y.** et al. (2024). *The bright black hole X-ray binary 4U 1543–47 during 2021 outburst - I. A clear state transition from super-Eddington to sub-Eddington accretion revealed by Insight-HXMT*. Monthly Notices of the Royal Astronomical Society, 530(1), 929-946.
- [6] Svoboda, J., Dovciak, M., Steiner, J. et al., **incl. Zhang, Y.** (2024). *Dramatic drop in the X-ray polarization of Swift J1727.8–1613 in the soft spectral state*. The Astrophysical Journal Letters, 966(2), L35.
- [7] Zhao, S., Tao, L., Li, P. et al., **incl. Zhang, Y.** (2024). *The bright black hole X-ray binary 4U 1543–47 during 2021 outburst: a thick accretion disk inflated by high luminosity*. Astronomy & Astrophysics, 685, A42.
- [8] Zhang, L., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2023). *Type-A quasi-periodic oscillation in the black hole transient MAXI J1348–630*. Monthly Notices of the Royal

Astronomical Society, 526(3), 3944-3950.

- [9] Ma, R, Méndez, M., García, F. et al., **incl. Zhang, Y.** (2023). *Variable corona during the transition from type-C to type-B quasi-periodic oscillations in the black hole X-ray binary MAXI J1820+070*. Monthly Notices of the Royal Astronomical Society, 525(1), 854-875.
- [10] Yang, Z. X., Zhang, L., Zhang S. N. et al., **incl. Zhang, Y.** (2023). *Fast transitions of X-ray variability in the black hole transient GX 339–4: comparison with MAXI J1820+070 and MAXI J1348–630*. Monthly Notices of the Royal Astronomical Society, 521(3), 3570-3584.
- [11] García, F., Karpouzas, K., Méndez, M. et al., **incl. Zhang, Y.** (2022). *The evolving properties of the corona of GRS 1915+105: a spectral-timing perspective through variable-Comptonization modelling*. Monthly Notices of the Royal Astronomical Society, 513(3), 4196-4207.
- [12] Liu, H., Fu, Y., Bambi, C. et al., **incl. Zhang, Y.** (2022). *The disk wind in GRS 1915+105 as seen by Insight-HXMT*. The Astrophysical Journal, 933(2), 122.
- [13] Yang, Z. X., Liang, Z., Bu, Q. C. et al., **incl. Zhang, Y.** (2022). *The accretion flow geometry of MAXI J1820+070 through broadband noise research with Insight-HXMT*. The Astrophysical Journal, 932(1), 7.
- [14] Méndez, M., Karpouzas, K., García, F. et al., **incl. Zhang, Y.** (2022). *Coupling between the accreting corona and the relativistic jet in the microquasar GRS 1915+105*. Nature Astronomy, 6(5), 577-583.
- [15] Karpouzas, K., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2021). *A variable corona for GRS 1915+105*. Monthly Notices of the Royal Astronomical Society, 503(4), 5522-5533.
- [16] Abdikamalov, A. B., Ayzenberg, D., Bambi, C. et al. **incl. Zhang, Y.** (2021). *Implementation of a radial disk ionization profile in the relxill_nk model*. Physical Review D, 103(10), 103023.
- [17] Liu, H., Ji, L., Bambi, C. et al., **incl. Zhang, Y.** (2021). *Testing evolution of LFQ-POs with mass accretion rate in GRS 1915+105 with Insight-HXMT*. The Astrophysical Journal, 909(1), 63.